EE2004: Digital Signal Processing

Tutorial 5
(March 2, 2026)

1. Using partial fraction expansion find the inverse z-transform of the following:
   (a) Find the $x[n] \in \ell_1$ whose z-transform is

$$X(z) = \frac{1 - 2z^{-1}}{1 - \frac{5}{2}z^{-1} + z^{-2}}$$

   (b) Find $x[n]$ whose z-transform is

$$X(z) = \frac{1}{(1 - \frac{1}{4}z^{-1})(1 - \frac{1}{2}z^{-1})^3} \qquad |z| > \frac{1}{2}$$

2. Using series expansion find the inverse z-transform of

$$X(z) = \frac{1 + 2z^{-1}}{1 - 2z^{-1} + z^{-2}}$$

   when $x[n]$ is (a) causal, and (b) non-causal.

3. Consider

$$Y(z) = \frac{1 - a^2}{1 + a^2 - a(z + z^{-1})}$$

   Using long-division, find (i) causal inverse z-transform (quotient should be in powers of z^{-1}), and (ii) anti-causal inverse z-transform (quotient should be in powers of z).

4. The input to a causal LTI system is $x[n] = 2^n\, u[-n-1] + (1/2)^n u[n]$. The output z-transform is

$$Y(z) = \frac{-\frac{3}{2}z^{-1}}{\left(1 + \frac{1}{2}z^{-1}\right)(1 - 2\,z^{-1})}$$

   (a) Determine $H(z)$. Be sure to specify the RoC.
   (b) What is the RoC for $Y(z)$?
   (c) Determine $y[n]$.

5. The following facts are given about and LTI system characterized by $h[n] \overset{\mathcal{Z}}{\longleftrightarrow} H(z)$: (i) $h[n]$ is real-valued, (ii) $h[n]$ is right-sided, (iii) $\lim_{z\to\infty} H(z) = 1$, (iv) $H(z)$ has two zeros, and (v) $H(z)$ has a complex-valued pole on the circle defined by $|z| = 3/4$. Is the system causal? Is it stable?

6. Using the contour integral method, find $x[n]$ for all n when $X(z) = \dfrac{1}{1 - a\,z^{-1}}$ with RoC $|z| < |a|$.

7. Find the sequences whose DTFT are given below: (a) $X(e^{j\omega}) = (\cos\omega)^N$ for an arbitrary integer $N > 0$, and (b) $X(e^{j\omega}) = (\cos\frac{\omega}{2})^N$ for an arbitrary *even* integer $N > 0$. *Hint: Express the cosine function in terms of complex exponentials and apply the binomial theorem.*

8. Consider the DTFT $H(e^{j\omega}) = j\omega$ for $\omega \in [-\pi, \pi)$. (a) Determine the corresponding $h[n]$. What is the value of $h[0]$? (b) Apply Parseval's theorem to this transform pair to obtain the value of $\displaystyle\sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} \frac{1}{n^2}$.

9. Consider the sequence $x[n] = \{2, 3, \underset{\downarrow}{-1}, 0, -4, 3, 1, 2, 4\}$. Find the inverse DTFT of (a) $X(e^{-j\omega})$, and (b) $X(-e^{j\omega})$. Comment on the difference between the two answers.

10. What symmetry property, if any, does the DTFT possess in each of the following cases: (a) $x[n] \in \mathbb{R}$, (b) $x[n] = x[-n]$, in addition to being real-valued, and (c) $x[n] = x^*[-n]$.

11. For each of the following Fourier transforms, use DTFT properties to determine whether or not the corresponding time-domain signal is (i) real, imaginary, or neither, and (ii) even, odd, or neither. Do this without explicitly evaluating the inverse of any of the given transforms.

    (a) $X_1(e^{j\omega}) = e^{-j\omega} \displaystyle\sum_{k=1}^{10} \sin(k\omega)$

    (b) $X_2(e^{j\omega}) = j\sin(\omega)\cos(5\omega)$

    (c) $X_3(e^{j\omega}) = A(\omega) + e^{jB(\omega)}$ where $A(\omega) = \begin{cases} 1, & 0 \le |\omega| \le \frac{\pi}{8} \\ 0, & \frac{\pi}{8} < |\omega| \le \pi \end{cases}$ and $B(\omega) = -\dfrac{3\omega}{2} + \pi$

12. Determine $x[n]$ given the following four facts about its DTFT $X(e^{j\omega})$:

    1. $x[n] = 0$ for $n > 0$
    2. $x[0] > 0$
    3. $\mathrm{Im}\{X(e^{j\omega})\} = \sin(\omega) - \sin(2\omega)$
    4. $\dfrac{1}{2\pi}\displaystyle\int_{-\pi}^{\pi} |X(e^{j\omega})|^2\, d\omega = 3.$

13. Using DTFT properties determine the numerical value of $A = \displaystyle\sum_{n=0}^{\infty} n\left(\frac{1}{2}\right)^n$. ◇